

STA623 - Bayesian Data Analysis

Take-home assignment: in-hospital mortality from severe pneumonia

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2026-10-02

Weight	25% of the overall course grade
Released	2 October 2026
Report due	2 December 2026, 23:59 CAT
Submit	One PDF report and a single Quarto (.qmd) or R script (.R) file that produces the report / the results; email these documents to <code>mhenrion@mlw.mw</code>
Oral examination	A 15-minute oral on your submitted report, 1–2 weeks after the due date: a separate assessment worth a further 25%

1. Overview

This is a computational project. You will **fit Bayesian regression models for in-hospital mortality** among children admitted with severe pneumonia. The outcome is whether each child died in hospital (**died**: 1 = died, 0 = discharged alive). You will model it with **logistic regression**, as a function of the child's characteristics at admission (oxygen saturation, nutritional status, age, sex) and of the hospital the child was admitted to (a **hospital random intercept**, plus hospital-level characteristics). All models are fitted in **NIMBLE**.

Around this model you will carry out a complete Bayesian analysis of your own dataset: checking the data, choosing and checking priors, fitting and checking the models, and using them to answer three questions that a district health management team (DHMT) needs answered.

The emphasis is on **your reasoning**. Most of the marks are for explaining *why* you made each choice, what the alternatives were, and what your results mean, not for producing a particular number. There is rarely a single correct answer; there are well-argued and poorly argued ones.

Your dataset is unique to you. Every student receives a different dataset, generated with its own parameter values. Your results will not match anyone else's, and the problems you meet in your data will differ from those of your classmates.

2. Background

Pneumonia is a leading cause of death among children under five in Malawi. Children admitted to hospital with severe pneumonia are assessed at admission for, among other things, **hypoxaemia** (low blood oxygen saturation, measured with a pulse oximeter) and **severe acute malnutrition**. Both are thought to increase the risk of death. Oxygen therapy treats hypoxaemia, but not every hospital has a functioning oxygen concentrator on the paediatric ward.

You have been given data on children aged 2–59 months admitted with severe pneumonia to 12–16 hospitals over one year. Most hospitals are district hospitals; one or two are central (referral) hospitals. The data are **as received from the hospitals and have not been cleaned**.

The data are simulated to resemble a real study. Hospitals are anonymised and the results must not be interpreted as describing real hospitals.

3. Your data

You receive two files, named with your student ID.

STA623_assignment_<ID>_children.csv: one row per admitted child.

Variable	Description
child_id	Anonymised child identifier
hospital	Hospital code (links to the hospitals file)
age_months or age_days	Age at admission; the unit is given in the column name
sex	F or M
spo2	Peripheral oxygen saturation at admission (%), measured on room air
muac_mm	Mid-upper arm circumference at admission, in millimetres
died	1 = died in hospital, 0 = discharged alive

STA623_assignment_<ID>_hospitals.csv: one row per hospital.

Variable	Description
hospital	Hospital code
level	district or central
oxygen_concentrator	yes if a functioning oxygen concentrator was available on the paediatric ward throughout the year

Standard clinical definitions: **hypoxaemia** is $\text{SpO}_2 < 90\%$; **severe acute malnutrition (SAM)** is $\text{MUAC} < 115 \text{ mm}$.

4. Research questions

RQ1. How much higher is the risk of in-hospital death for children who are hypoxaemic at admission, after accounting for other factors? Report this both as an odds ratio and in terms the DHMT can understand (for example, deaths per 100 children).

RQ2. Is the availability of an oxygen concentrator associated with lower in-hospital mortality? Be explicit about what your data can and cannot tell you about this.

RQ3. The DHMT can fund supportive supervision visits to **two of the district hospitals** next quarter. Based on mortality after accounting for differences in the children each hospital admits, which two would you recommend? State the criterion you used to decide, how confident you are, and what you would advise for hospitals where the data are too sparse to tell.

5. Tasks

Organise your report around the parts below. The marks for each part are shown in brackets (total 100).

Part A: Data checking and preparation [20]

- Import and explore the data. Check every variable for problems: impossible or implausible values, coding problems, missing values, inconsistencies between hospitals.
- For **each problem you find**, report how you found it (show the relevant table or plot), how you handled it, and why. If a choice affects which children are analysed, say how many and what assumption your choice makes.
- Decide how to enter age, oxygen saturation and nutritional status into your models (for example: continuous or categorised; which scale; centred or not). Justify each decision.
- Summarise the data by hospital (admissions, deaths, hypoxaemia, SAM).

Part B: Priors and prior predictive checks [10]

1. **Priors for the key parameters.** For the intercept (baseline mortality), the hypoxaemia effect and the between-hospital standard deviation, choose priors that reflect what is clinically plausible. State what you assumed (for example: a plausible range of mortality for a typical child; a plausible size for the effect of hypoxaemia; how much district hospitals could plausibly differ), where that assumption comes from (cite any source you use), and show every step of translating it into a distribution on the scale of the parameter.
2. Choose **weakly informative priors** for all other parameters, and justify them.
3. **Prior predictive checks:** simulate from your priors and show what they imply for (i) mortality of a typical child at an average hospital, (ii) mortality of a hypoxaemic child, and (iii) the spread of mortality between hospitals.
4. If the checks show implausible implications, revise your priors and check again. Report both the original and the revised priors.

Part C: Models and computation [20]

- Your models have the general form

$$\text{died}_i \mid p_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \beta_0 + \mathbf{x}_i^\top \beta + u_{h(i)}, \quad u_j \sim \mathcal{N}(0, \sigma^2),$$

where $\mathbf{x}_i$ are the child-level covariates of child i , $h(i)$ is the hospital that admitted child i , and u_j is the random intercept of hospital j . Which child-level covariates to include, and how to code them, is your decision (Part A).

- Fit, using NIMBLE, at least the following two models:
 - **M1:** child-level covariates and hospital random intercepts, as above;
 - **M2:** M1 plus the hospital-level covariates `level` and `oxygen_concentrator` in the linear predictor.

You may fit additional models (for example with interactions or a different form for age) if your reasoning or your model checks suggest them.

- Write each model in full mathematical notation, including the priors, as well as in code. The main models must be fitted with your own NIMBLE code; you may reuse code from the course materials.
- For **every model you interpret**, report convergence diagnostics: R-hat, bulk and tail ESS, and trace plots for the key parameters. State the criteria you used.
- **Computational problem:** describe at least one computational problem you encountered during the assignment. Show the evidence from the problematic run (diagnostic table or trace plot), explain what caused it, and how you fixed it. If you genuinely met no problems, identify which problem you would expect with these data, provoke it deliberately, and show that your approach avoids it.

Part D: Model checking and comparison [20]

- Carry out **posterior predictive checks** for your main model using at least **two test statistics you have chosen**. At least one must target a specific model assumption you had reason to doubt. For each, explain why you chose it, show the result (plot and posterior predictive p-value), and interpret it.
- If a check reveals a problem, act on it: revise the model and check again.
- **Compare your models using LOO.** Explain what the comparison tells you, and whether and why it changes your choice of final model. A number alone is not an answer.

Part E: Answers to the research questions [20]

- Answer RQ1–RQ3 with posterior summaries on interpretable scales: posterior medians, credible intervals and posterior probabilities of relevant events.
- For RQ3, make an explicit recommendation and explain the decision criterion (for example: probability of excess mortality, size of the excess, expected excess deaths, or a combination). Discuss the smallest hospitals explicitly.

Part F: Sensitivity to the priors [5]

- Refit your final model with your priors replaced by a clearly different set (for example much vaguer priors, as often seen in older BUGS code).
- Report how, and how much, your answers to RQ1–RQ3 change. Which conclusions are robust and which depend on the priors? Explain why.

Part G: Decision log, reproducibility and AI-use statement [5]

- **Decision log (Appendix A of your report):** a table with **at least 6 entries**, written *as you go* rather than at the end, with columns: number, date, decision, alternatives considered, choice and reason, what evidence would make you change your mind.
- **AI-use statement (Appendix B):** see Section 7.
- **Reproducibility:** your `.qmd` file must render your complete report from your two data files, placed in the same folder, with no manual steps other than installing packages. Set random seeds. Every number in the text must come from the rendered code (use inline R code, not typed-in numbers). End the report with `sessionInfo()`.

6. Report format

- One PDF with your results (either rendered from a Quarto file or a Word document saved to pdf). Main text (Parts A–F): **at most 10 pages**, including figures and tables. Appendices (decision log, AI statement, `sessionInfo()`) do not count towards the limit.
- Show the code for your models and key computations. Long or repetitive code can be hidden in the PDF with `#| echo: false` (if you use Quarto), but it must be in the `.qmd` or `.R` file.
- Write in your own words. Clear, direct English is expected; polished prose is not.

7. Use of AI tools and academic integrity

- You **may** use AI tools (such as ChatGPT or Claude) to help with coding, debugging, explanations of concepts and language, **provided you declare it**. You are fully responsible for everything you submit, including any errors an AI tool introduced.
- **Appendix B (AI-use statement)** must list which tools you used and what for, give examples of your prompts, and describe **at least one instance where an AI tool's output was wrong, misleading or rejected by you**, and how you knew. If you did not use AI tools, state this.
- You will be **examined orally on your submitted report**, including any code an AI tool helped you write. Make sure you understand, and can explain, everything you submit.
- You may discuss concepts with classmates, but your code, analysis and report must be your own. Do not share your data or code. Reports with shared code or wording will be investigated.

8. Marking summary

Component	Marks
A. Data checking and preparation	20
B. Priors and prior predictive checks	10
C. Models and computation	20
D. Model checking and comparison	20
E. Answers to the research questions	20
F. Sensitivity to the priors	5
G. Decision log, reproducibility, AI statement	5
Total	100

In every part, the explanation of your reasoning counts for more than the result. A correct model with no justification earns fewer marks than a reasonable model with clear, honest reasoning about its limitations. Your reasoning must refer to **your own data**: your numbers, plots and problems. Generic statements that could apply to any dataset (“the model converged well”, “the checks showed a good fit”) earn limited marks, however well written.

9. Practical advice

- **Start early.** Aim to have your data checked and a first model running by the end of the first week: fitting, checking and revising models takes longer than you expect.
- **NIMBLE** models with a few thousand children take a few minutes to compile and run. During development, use short runs and cache chunks (`#| cache: true`); render the final report from scratch.
- Lectures 4 and 5 and Practicals 4 and 5 contain most of the tools you need. Posterior predictive checks and LOO are in Lecture 5; LOO needs the pointwise log-likelihood, which you can compute from your posterior draws as shown there.
- If your sampler struggles, think about *why* before running it longer (Lecture 4).
- Keep your decision log open while you work, and add to it whenever you choose between alternatives.